Exercise 3 Solutions

1. $X_t = \frac{X_{t-1}}{12} + \frac{X_{t-2}}{12} + Z_t$, where Z_t has mean 0 and variance σ^2

$E(X_t) = \frac{1}{12}E(X_{t-1}) + \frac{1}{12}E(X_{t-2}) + 0 \Rightarrow \mu = \frac{1}{6}\mu \Rightarrow \mu = 0$ free of t.

$V(X_t) = (\frac{1}{12})^2 V(X_{t-1}) + (\frac{1}{12})^2 V(X_{t-2}) + 2(\frac{1}{12})(\frac{1}{12})E(X_{t-1}X_{t-2}) + V(Z_t)$

$\sigma_X^2 = \frac{\sigma_X^2}{144} + \frac{\sigma_X^2}{144} + \frac{1}{72}\gamma(1) + \sigma^2 = \frac{\sigma_X^2}{72} + \frac{\sigma_X^2}{72(11)} + \sigma^2 \Rightarrow \sigma_X^2 = \frac{66}{65}\sigma^2$ free of t.

$Cov(X_t, X_{t-k}) = E(X_t X_{t-k}) = E(\frac{1}{12}X_{t-1}X_{t-k} + \frac{1}{12}X_{t-2}X_{t-k} + Z_t X_{t-k})$

$\gamma(k) = \frac{1}{12}\gamma(k-1) + \frac{1}{12}\gamma(k-2)$ free of t.

$k=1 \Rightarrow \gamma(1) = \frac{1}{12}\gamma(0) + \frac{1}{12}\gamma(1)$

$\gamma(1) = \frac{\sigma_X^2}{12} + \frac{1}{12}\gamma(1) \Rightarrow \gamma(1) = \frac{\sigma_X^2}{11}$

$\therefore X_t$ is stationary.

$\rho(k) = \frac{1}{12}\rho(k-1) + \frac{1}{12}\rho(k-2)$

N.B. $\rho(-k) = \rho(k)$, $\rho(0) = 1$

$\rho(1) = \frac{1}{12}\rho(0) + \frac{1}{12}\rho(1) = \frac{1}{12} + \frac{\rho(1)}{12}$

$\therefore \rho(1) = \frac{\frac{1}{12}}{1-\frac{1}{12}} = \frac{1}{11}$

Then $X_t = k_1(\frac{1}{\lambda_1})^k + k_2(\frac{1}{\lambda_2})^k$ (see p.9)

where λ_1 and λ_2 are the roots of

$1 - \frac{B}{12} - \frac{B^2}{12} = 0 \Rightarrow 12 - B - B^2 = (B+4)(B-3) = 0$

$\therefore B = \begin{cases} 3 & r_1 = \frac{1}{\lambda_1} = \frac{1}{3} \\ -4 & r_2 = \frac{1}{\lambda_2} = -\frac{1}{4} \end{cases}$

Then $\rho(0) = k_1 + k_2 = 1$ —— ①

$\rho(1) = \frac{k_1}{3} - \frac{k_2}{4} = \frac{1}{11} \Rightarrow 4k_1 - 3k_2 = \frac{12}{11}$ —— ②

$4 \times ① - ②$

$4k_1 + 4k_2 = 4$

$-) \; 4k_1 - 3k_2 = \frac{12}{11}$

$7k_2 = 4 - \frac{12}{11} = \frac{44-12}{11} \Rightarrow k_2 = \frac{32}{77}$

Put $k_2 = \frac{32}{77}$ into ① $\Rightarrow k_1 = 1 - \frac{32}{77} = \frac{45}{77}$.

$\therefore \rho(k) = \frac{45}{77}(\frac{1}{3})^k + \frac{32}{77}(-\frac{1}{4})^k$, $k = 0, 1, 2, \ldots$

2. $X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + Z_t$ (similar to q. 1)

$\Rightarrow \rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2)$

$\rho(1) = \alpha_1 \rho(0) + \alpha_2 \rho(1) = \alpha_1 + \alpha_2 \rho(1)$

$\rho(1) = \frac{\alpha_1}{1-\alpha_2} = \begin{cases} (0.3,\ 0.5) \\ (0.3,\ -0.5) \end{cases} \Rightarrow \begin{cases} \frac{0.3}{1-0.5} = \frac{0.3}{0.5} = 0.6 \\ \frac{0.3}{1+0.5} = \frac{0.3}{1.5} = 0.2 \end{cases}$

$\rho(2) = \alpha_1 \rho(1) + \alpha_2 \rho(0) = \begin{cases} 0.3(0.6) + 0.5 = 0.68 \\ 0.3(0.2) - 0.5 = -0.44 \end{cases}$

$\rho(3) = \alpha_1 \rho(2) + \alpha_2 \rho(1) = \begin{cases} 0.3(0.68) + 0.5(0.6) = 0.504 \\ 0.3(-0.44) - 0.5(0.2) = -0.232 \end{cases}$

$\rho(4) = \alpha_1 \rho(3) + \alpha_2 \rho(2) = \begin{cases} 0.3(0.504) + 0.5(0.68) = 0.4912 \\ 0.3(-0.232) - 0.5(-0.44) = 0.1504 \end{cases}$

$\rho(5) = \alpha_1 \rho(4) + \alpha_2 \rho(3) = \begin{cases} 0.3(0.4912) + 0.5(0.504) = 0.3994 \simeq 0.40 \\ 0.3(0.1504) - 0.5(-0.232) = 0.16112 \simeq 0.16 \end{cases}$

$\rho(6) = \alpha_1 \rho(5) + \alpha_2 \rho(4) = \begin{cases} 0.3(0.40) + 0.5(0.4912) = 0.3656 \simeq 0.37 \\ 0.3(0.16112) - 0.5(0.1504) = -0.0269 \simeq -0.027 \end{cases}$

$\rho(7) = \alpha_1 \rho(6) + \alpha_2 \rho(5) = \begin{cases} 0.3(0.3656) + 0.5(0.40) = 0.30968 \simeq 0.31 \\ 0.3(-0.0269) - 0.5(0.16112) = -0.0886 \simeq -0.09 \end{cases}$

$\rho(8) = \alpha_1 \rho(7) + \alpha_2 \rho(6) = \begin{cases} 0.3(0.30968) + 0.5(0.3656) = 0.2757 \simeq 0.28 \\ 0.3(-0.0886) - 0.5(-0.0269) = -0.01313 \simeq -0.013 \end{cases}$

$\rho(9) = \alpha_1 \rho(8) + \alpha_2 \rho(7) = \begin{cases} 0.3(0.2757) + 0.5(0.30968) = 0.2376 \simeq 0.24 \\ 0.3(-0.01313) - 0.5(-0.0886) = 0.0404 \simeq 0.04 \end{cases}$

$\rho(10) = \alpha_1 \rho(9) + \alpha_2 \rho(8) = \begin{cases} 0.3(0.2376) + 0.5(0.2757) = 0.2091 \simeq 0.21 \\ 0.3(0.0404) - 0.5(-0.01313) = 0.0187 \simeq 0.019 \end{cases}$

3. $X_t = A\cos(\omega t) + B\sin(\omega t), \quad t = 0, 1, 2, \ldots$ A, B uncorrelated r.v.s; ω constant.

$E(X_t) = E[A\cos(\omega t) + B\sin(\omega t)] = E(A)\cos(\omega t) + E(B)\sin(\omega t) = 0$ free of t

$V(X_t) = V[A\cos(\omega t) + B\sin(\omega t)] = \cos^2(\omega t)V(A) + \sin^2(\omega t)V(B)$ free of t

$= \cos^2(\omega t) + \sin^2(\omega t) = 1$

$X_t X_{t-k} = [A\cos(\omega t) + B\sin(\omega t)][A\cos(\omega t - \omega k) + B\sin(\omega t - \omega k)]$

$= A^2\cos(\omega t)\cos(\omega t - \omega k) + AB\sin(\omega t)\cos(\omega t - \omega k) + AB\cos(\omega t)\sin(\omega t - \omega k) + B^2\sin(\omega t)\sin(\omega t - \omega k)$

$E(X_t X_{t-k}) = \cos(\omega t)\cos(\omega t - \omega k) + \sin(\omega t)\sin(\omega t - \omega k) = \cos[\omega t - (\omega t - \omega k)]$

$= \cos(\omega k)$ free of t

N.B. $\cos(C - D) = \cos C \cos D + \sin C \sin D$ where $C = \omega t$, $D = \omega t - \omega k$

$\therefore X_t$ is stationary

4. $y_t = 12 + 0.4\,y_{t-1} + 0.1\,y_{t-2} + \varepsilon_t + 0.3\varepsilon_{t-1} - 0.2\,\varepsilon_{t-2}$

$E(y_t) = 12 + 0.4E(y_{t-1}) + 0.1\,E(y_{t-2}) + 0$

$\mu = 12 + 0.4\mu + 0.1\mu \Rightarrow 0.5\mu = 12 \Rightarrow \mu = 24.$

5. $Z_t = 5 + a_t - 0.5a_{t-1} + 0.25a_{t-2} \Rightarrow \beta_0 = 1,\ \beta_1 = -0.5 = -\frac{1}{2},\ \beta_2 = 0.25 = \frac{1}{4}$

$m = 2$

$$\rho(k) = \frac{\sum_{i=0}^{m-k} \beta_i \beta_{i+k}}{\sum_{i=0}^{m} \beta_i^2} \quad (\text{p. 3 bottom})$$

$$\rho(1) = \frac{\beta_0\beta_1 + \beta_1\beta_2}{\beta_0^2 + \beta_1^2 + \beta_2^2} = \frac{-\frac{1}{2} + (-\frac{1}{2})(\frac{1}{4})}{1 + (-\frac{1}{2})^2 + (\frac{1}{4})^2} = \frac{-\frac{1}{2} - \frac{1}{8}}{1 + \frac{1}{4} + \frac{1}{16}} = \frac{-\frac{5}{8}}{\frac{21}{16}} = -\frac{10}{21}$$

$$\rho(2) = \frac{\beta_0\beta_2}{\frac{21}{16}} = \frac{\frac{1}{4}}{\frac{21}{16}} = \frac{4}{21}$$

$\rho(k) = 0$, for $k \geq 3$.

6. $Z_t = 0.8Z_{t-1} + a_t + 0.7a_{t-1} + 0.6a_{t-2}$

$Z_t Z_{t-k} = 0.8Z_{t-1}Z_{t-k} + a_t Z_{t-k} + 0.7a_{t-1}Z_{t-k} + 0.6a_{t-2}Z_{t-k}$

$\gamma(k) = E(Z_t Z_{t-k}) = 0.8\,E(Z_{t-1}Z_{t-k}) = 0.8\,\gamma(k-1)$

$\rho(k) = \frac{\gamma(k)}{\gamma(0)} \Rightarrow \rho(k) = 0.8\,\rho(k-1)$, for $k \geq 3$.

For $k = 2$, then $\gamma(2) = 0.8\,\gamma(1) + 0.6\,E(a_{t-2}Z_{t-2})$

$E(a_{t-2}Z_{t-2}) = E[a_{t-2}(0.8Z_{t-3} + a_{t-2} + 0.7a_{t-3} + 0.6a_{t-4})] = E(a_{t-2}^2) = \sigma_a^2$

$\rho(2) = 0.8\,\rho(1) + 0.6\,\frac{\sigma_a^2}{\gamma_0}$

$\gamma(1) = E(Z_t Z_{t-1}) = 0.8\,E(Z_{t-1}^2) + E(a_t Z_{t-1}) + 0.7E(a_{t-1}Z_{t-1}) + 0.6E(a_{t-2}Z_{t-1})$

$E(a_t Z_{t-1}) = E[a_t(0.8Z_{t-2} + a_{t-1} + 0.7a_{t-2} + 0.6a_{t-3})] = 0$

$E(a_{t-1}Z_{t-1}) = E[a_{t-1}(0.8Z_{t-2} + a_{t-1} + 0.7a_{t-2} + 0.6a_{t-3})] = \sigma_a^2$

$E(a_{t-2}Z_{t-1}) = E[a_{t-2}(0.8Z_{t-2} + a_{t-1} + 0.7a_{t-2} + 0.6a_{t-3})] = 0.8\sigma_a^2 + 0.7\sigma_a^2 = 1.5\sigma_a^2$

$\gamma(1) = 0.8\gamma_0 + 0 + 0.7\sigma_a^2 + 0.6(1.5\sigma_a^2)$

$= 0.8\gamma_0 + 1.6\sigma_a^2$

$\therefore \rho(1) = 0.8 + 1.6\,\frac{\sigma_a^2}{\gamma_0} \Rightarrow \rho(2) = 0.64 + 1.28\,\frac{\sigma_a^2}{\gamma_0} + 0.6\,\frac{\sigma_a^2}{\gamma_0}$

$= 0.64 + 1.88\,\frac{\sigma_a^2}{\gamma_0}$

7. $y_t = 0.3\, y_{t-1} + 0.2\, y_{t-2} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}$

$\rho(1) = 0.8, \quad \rho(2) = -0.2$

$E(y_t y_{t-k}) = E\left[0.3 y_{t-1} y_{t-k} + 0.2 y_{t-2} y_{t-k} + \varepsilon_t y_{t-k} - \theta_1 \varepsilon_{t-1} y_{t-k} - \theta_2 \varepsilon_{t-2} y_{t-k}\right]$

$\rho(k) = 0.3\,\rho(k-1) + 0.2\,\rho(k-2)$

$\rho(3) = 0.3\,\rho(2) + 0.2\,\rho(1)$

$= 0.3(-0.2) + 0.2(0.8)$

$= -0.06 + 0.16$

$= 0.1$

$\rho(4) = 0.3\,\rho(3) + 0.2\,\rho(2)$

$= 0.3(0.1) + 0.2(-0.2)$

$= 0.03 - 0.04$

$= -0.01$

Since $y_t = 0.3\, y_{t-1} + 0.2\, y_{t-2} \Rightarrow 1 - 0.3B - 0.2B^2 = 0$

hence $2B^2 + 3B - 10 = 0 \Rightarrow |B| = \left|\frac{-3 \pm \sqrt{89}}{4}\right| > 1.$

Thus, y_t has parameter values (0.3 and 0.2) all less than 1 and the roots of y_t lie outside the unit circle. Hence y_t is stationary.